

Ivanna Aleman-Coronado

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EDUCATION

The University of Texas at Austin – Austin, TX | Robotics Honors Program
B.S. Computer Science, Minor in Robotics

May 2029

- Relevant Coursework: Data Structures, Computer Architecture, Operating Systems, Autonomous Driving, Discrete Math, Linear Algebra, Probability & Statistics

TECHNICAL SKILLS

Languages: Python, Java, C, JavaScript, HTML/CSS, Bash, AArch64 Assembly

Frameworks & Libraries: ROS 2, React, Vite, NumPy, Pandas, PufferLib, librosa

Developer Tools: Git, Linux/Unix, GDB, Make, VS Code, PyCharm, Jupyter, Figma, Foxglove

Systems & Infrastructure: Distributed training on HPC clusters (TACC), Boston Dynamics Spot SDK, Unity, Netlify

ML/AI: Reinforcement Learning (PPO), Imitation Learning, LLM-based evaluation, K-means/GMM clustering

Languages (Spoken): English (Fluent), Spanish (Fluent)

EXPERIENCE

Undergraduate Research Fellow, Autonomous Mobile Robotics Laboratory (AMRL)
Austin, Texas

Jan 2026 - Present

- Engineered an end-to-end ROS 2 system in Python that drives a Boston Dynamics Spot quadruped through a fully autonomous, music-synchronized 50-second routine
- Built a real-time beat-detection module with librosa that extracts tempo and beat times from audio and schedules motion commands.
- Designed a modular move library and choreography state machine so new routines compose from reusable primitives.
- Debugged low-level body-pose control failures in the ros2 driver to restore reliable, precisely timed motion-command execution
- Adopted as a recurring lab showcase and educational demo of the sense-plan-act loop: audio sensing (beat detection), planning (choreography state machine), and actuation (timed Spot motion)

Machine Learning Research Intern, University of Texas at Dallas (StaRLiNG Lab)
Dallas, Texas

May 2025 - August 2025

- Built an end-to-end Python data pipeline that ingests raw network traffic captures, cleans and transforms them with Pandas, and computes feature vectors with NumPy for downstream models
- Implemented modular K-means, Gaussian Mixture Model, and One-Class classifier components behind a shared interface, so detection methods could be swapped and compared without changing pipeline code
- Built an evaluation harness that scores anomaly-based against signature-based detection on accuracy, precision, and recall; anomaly-based models outperformed signature baselines on novel attack patterns

PROJECTS

Co-Adaptive Value Alignment for Autonomous Vehicles

Python, Unity, PPO, PufferLib, TACC HPC, Qwen2.5-14B

- Designed a multi-agent reinforcement learning environment in Unity simulating coadaptive merging behavior between human and autonomous drivers, training PPO policies at scale on TACC HPC clusters via PufferLib.
- Integrated a Qwen2.5-14B LLM as a behavior-scoring critic to evaluate driving safety/courtesy in dynamic race-track scenarios, feeding reward signals back into the multi-agent decision-making framework.

Pipelined Processor Emulator & Cache Simulator

C, AArch64 Assembly

- Designed and implemented a cycle-accurate emulator in C for a pipelined AArch64 processor, modeling instruction fetch, decode, execute, memory, and writeback stages
- Implemented hazard detection and resolution logic (data forwarding and pipeline stalls) to preserve correct execution across dependent instructions.
- Built a configurable set-associative cache simulator that replays memory access traces under varying set, associativity, and block-size parameters, with LRU replacement and hit/miss/eviction accounting

LEADERSHIP & AWARDS

- **Scholar, Hispanic Scholarship Fund (HSF)** — Recognized for academic achievement and leadership.
- **Social Officer, The Hispanic Association of Computer Scientists** — Plan and execute social events for 50+ members to build community among Hispanic students in CS and foster peer support and retention in the major
- **Student Ambassador, Texas Exes** — represents the UT Austin alumni association on campus, connecting students to the Texas Exes network (HookedIn, alumni mentorship) and promoting membership benefits and resources to strengthen student-alumni relationships.
- **Mentor, Robotics Honors Program** — mentor incoming students in the UT Austin Robotics Honors Program, advising on coursework, research opportunities, and navigating the program